QUANT

Mensuration & Geometry

(Triangles, Polygons & Mensuration)

Assignment 04

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Geometry Online Assignment Cluster I -04

1. A rectangular garden is 25 meter long & 15 meter wide. A 2 meter wide path is made all around it on the inside. Find the area of path in sq. meter.

1. 144 2. 176 3. 216 4. None of these

2. A square tin sheet of side 12 inches is converted into a box with open top in the following steps: The sheet is placed horizontally. Then, equal sized sqares, each of side x inches, are cut from the four corners of the sheet. Finally, the four resulting sides are bent vertically upwards in the shape of a box. If x is an integer, then what value of x maximizes the volume of the box?

1. 3 2. 4 3. 1 4. 2

3. Euclid has just created a triangle whose longest side is 20. If the length of the other side is 10 cm and the area of the triangle is 80 sq. cm, then what is the length of the third side?

1. $\sqrt{260}$ 2. $\sqrt{250}$ 3. $\sqrt{240}$ 4. $\sqrt{270}$

4. The diameter of a circle is 21 meter. It will take how many revolutions to cover a distance of 6.6 km?

1. 80 2. 50 3. 200 4. 100

5. Find the area of quadrant of a circle, whose circumference is 44 cm (in cm^2):

1. 77 2. 38.5 3. 19.25 4. 19.25π

6. The inner circumference of a circular track is 440 cm. The track is 14 cm wide. Find the diameter of the outer track of the circle:

1. 84 cm 2. 154 cm 3. 336 cm 4. 168 cm

7. If two sides of a triangle are 5 & 12 cm, which of the following can be the area of the given triangle?

1. 27.65 2. 31.25 3. 30.21 4. 32.50

8. ABCD is an isosceles trapezium with parallel sides AB & CD having lengths of 6 cm and 12 cm respectively. Also AD = 6 cm. An in-circle is drawn inside ABCD with centre O. The area of the triangle OCD is (in sq. cm)

1. $9\sqrt{3}$ 2. $27\sqrt{3}$ 3. $18\sqrt{2}$ 4. Data inadequate

9. A ladder leans against a vertical wall. The top of the ladder is 8 m above the ground. When the bottom of the ladder is moved 2 m farther away from the wall, the top of the ladder rests against the foot of the wall. What is the length of the ladder?

1. 10 m 2. 15 m 3. 20 m 4. 17 m

10. A semi-circle is drawn with AB as its diameter. From C, a point on AB, a line perpendicular to AB is drawn meeting the circumference of the semi-circle at D. Given that AC = 2 cm and CD = 6 cm, the area of the semi-circle (in sq. cm) will be :

1. 32π 2. 50π 3. 40.5π 4. 81π

Answer key & Explanations

Q. No.	Ans.	Exp.
1.	1	The area of the rectangle is 25 × 15 = 375 sq m. Now find the area of the smaller rectangle after excluding the path, which is given to be inside i.e. 21 × 11 = 231 sq m. The difference is the area of the path i.e. 375 – 231 = 144 sq m.
2.	4	Volume of the box = l × b × h = (12 – 2x) (12 – 2x) (x) Putting x = 1, 2, 3, 4, we get the maximum value of the above equation at = 2. So maximum volume = 128.
3.	1	Let third side be x. S = (20 + 10 + x) / 2 = (30 + x) / 2. Now apply hero's formula and equate it with 80 sq. cm. given and get = $(900 – x^2)(x^2 – 100)$ = 6400 × 4 × 4 = 102400. On checking options we get x = 260. Because (900 – 260) × 160 =102400.
4.	4	No. of revolutions = Distance/circumference. Distance = 6.6 × 1000 = 6600 metres. Circumference = 2 × 22/7 × 21/2 = 66 metres. No. of revolutions = 6600/66 = 100 revolutions.
5.	2	As 2 × 22/7 × r = 44, so r = 7. Hence area of the quadrant = ¼ × 22/7 × 7 × 7 = 38.5 cm^2.
6.	4	As 2 × 22/7 × r = 440, so r = 70 cm. Hence d = 140 cm and the track is 14 cm wide from both the sides and thus the diameter becomes 140 + 28 = 168
7.	1	The maximum possible area is when these two sides become perpendicular to each other. In that case the area will become 30 and in any case it has to be lesser than that, any given option less than 30 will be the answer.
8.	1	Because the trapezium is isosceles, the oblique sides must be equal. Taking AB as 6 and dropping perpendiculars to the base from the two ends just create a rectangle in the middle. It will also result in two triangles at the two corners. The base angles on both the sides are already equal and one of the angles is right angles. Thus the three angles are equal and as oblique sides are equal, the two triangles become congruent. Applying the Pythagoras theorem, calculate the length of the perpendiculars dropped i.e. $6^2 – 3^2 = \sqrt{27} \rightarrow 3\sqrt{3}$. The radius of the incircle will be half of that i.e. $3\sqrt{3}/2$. Now find the area of the triangle as asked in the question. It will be ½ × base × height $\Rightarrow$ ½ × 12 × $3\sqrt{3}/2$ $\Rightarrow 9\sqrt{3}$.
9.	4	Again read the question and analyze the height is given to be 8 and the hypotenuse is 2 metres longer than the base. Go to the Pythagorean triplets and you can guess the only possibility is 8, 15, and 17. Hence the length of the ladder is 17 metres.
10.	2	$\left(\sqrt{40}\right)^2 + \left(\sqrt{36+x^2}\right)^2 = (2+x)^2$, $40+36+x^2 = 4+x^2+4x$ $\Rightarrow 4x = 72 \Rightarrow x = 18$. Diameter = 20 $\Rightarrow$ Radius = 10cms. Area of Semicircle = $\frac{1}{2}\pi \times 10^2 = 50\pi$.